I am working on lampp php 8 by bitnami on aws, do you know this vps?

Memory updated

Yes, I'm familiar with LAMP PHP 8 by Bitnami on AWS. It's a pre-configured stack that includes Apache, MySQL (or MariaDB), and PHP, making it easier to deploy PHP applications on AWS. Are you facing any issues or need help with setup, configuration, or deployment?

Now I have to set two laravel projects on this vps.

First domain is: nafiulakbar.xyz with ip 3.109.132.234
Second one is subdomain backend.nafiulakbar.xyz with ip 3.109.132.234

Now I will set ssl by letsencrypt which is a free ssl and both domain and subdomain will be have the ssl. Now guide me the way how to set two domains with ssl.

Dont worry about the laravel project setup, I can handle it but I want to know how to setup both domains online

Memory updated

Here's how you can configure both `nafiulakbar.xyz` and `backend.nafiulakbar.xyz` with SSL using Let's Encrypt on your LAMP PHP 8 Bitnami AWS VPS.

Step 1: Set Up Virtual Hosts for Both Domains

First, you need to create virtual hosts for your primary domain and subdomain.

1. Create Virtual Host Config Files

Bitnami uses Apache, so navigate to the Apache configuration directory:

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```
cd /opt/bitnami/apache2/conf/vhosts/
```

Now, create two virtual host configuration files:

For nafiulakbar.xyz

Create a file:

bash Copy Edit

```
sudo nano nafiulakbar-vhost.conf
```

Add the following configuration:

apache Copy Edit

```
<VirtualHost *:80>
    ServerName nafiulakbar.xyz
    DocumentRoot "/opt/bitnami/htdocs/nafiulakbar"

    <Directory "/opt/bitnami/htdocs/nafiulakbar">
        AllowOverride All
        Require all granted
    </Directory>

    ErrorLog "logs/nafiulakbar-error_log"
    CustomLog "logs/nafiulakbar-access_log" common
</VirtualHost>
```

For backend.nafiulakbar.xyz

Create another file:

bash Copy Edit

```
sudo nano backend-nafiulakbar-vhost.conf
```

Add this configuration:

apache Copy Edit

```
<VirtualHost *:80>
    ServerName backend.nafiulakbar.xyz
    DocumentRoot "/opt/bitnami/htdocs/backend"

    <Directory "/opt/bitnami/htdocs/backend">
        AllowOverride All
        Require all granted
    </Directory>

    ErrorLog "logs/backend-error_log"
    CustomLog "logs/backend-access_log" common
</VirtualHost>
```

Save and exit (`CTRL+X`, then `Y` and `Enter`).

2. Restart Apache

Run the following command to restart Apache:

bash Copy Edit

```
sudo /opt/bitnami/ctlscript.sh restart apache
```

Step 2: Set Up Let's Encrypt SSL

Now, install and configure Let's Encrypt SSL for both domains.

1. Install Certbot (if not installed)

```bash
sudo apt update
sudo apt install certbot
```

2. Generate SSL Certificates

Run this command to generate an SSL certificate for both domains:

```bash
sudo certbot certonly --apache -d nafiulakbar.xyz -d backend.nafiulakbar.xyz
```

Follow the instructions and verify your domain ownership.

3. Configure SSL in Apache

After obtaining the SSL certificates, update your virtual host configurations.

Update `nafiulakbar-vhost-ssl.conf`

Create or edit the SSL config:

```bash
sudo nano nafiulakbar-vhost-ssl.conf
```

Add:

```apache
<VirtualHost *:443>
    ServerName nafiulakbar.xyz
    DocumentRoot "/opt/bitnami/htdocs/nafiulakbar"

    SSLEngine on
    SSLCertificateFile "/etc/letsencrypt/live/nafiulakbar.xyz/fullchain.pem"
    SSLCertificateKeyFile "/etc/letsencrypt/live/nafiulakbar.xyz/privkey.pem"

    <Directory "/opt/bitnami/htdocs/nafiulakbar">
        AllowOverride All
        Require all granted
    </Directory>

    ErrorLog "logs/nafiulakbar-ssl-error_log"
    CustomLog "logs/nafiulakbar-ssl-access_log" common
</VirtualHost>
```

Update `backend-nafiulakbar-vhost-ssl.conf`

Create or edit the SSL config:

```bash
sudo nano backend-nafiulakbar-vhost-ssl.conf
```

Add:

```apache
<VirtualHost *:443>
    ServerName backend.nafiulakbar.xyz
    DocumentRoot "/opt/bitnami/htdocs/backend"

    SSLEngine on
    SSLCertificateFile "/etc/letsencrypt/live/nafiulakbar.xyz/fullchain.pem"
    SSLCertificateKeyFile "/etc/letsencrypt/live/nafiulakbar.xyz/privkey.pem"

    <Directory "/opt/bitnami/htdocs/backend">
        AllowOverride All
        Require all granted
    </Directory>

    ErrorLog "logs/backend-ssl-error_log"
    CustomLog "logs/backend-ssl-access_log" common
</VirtualHost>
```

Save and exit.

4. Restart Apache

```bash
sudo /opt/bitnami/ctlscript.sh restart apache
```

5. Set Up Auto-Renewal

To renew SSL certificates automatically, add a cron job:

```bash
sudo crontab -e
```

Add this line at the bottom:

```bash
0 3 * * * certbot renew --quiet && sudo /opt/bitnami/ctlscript.sh restart apache
```

This will renew SSL certificates every 3 AM.

Step 3: Test Your Setup

- Visit `https://nafiulakbar.xyz` and `https://backend.nafiulakbar.xyz` to check if SSL is working.
- Run:

```bash
sudo apachectl -t
```

to check if Apache has any configuration errors.

Let me know if you face any issues! 🚀

After running this command it shows,

Saving debug log to /var/log/letsencrypt/letsencrypt.log
The requested apache plugin does not appear to be installed

The error **"The requested apache plugin does not appear to be installed"** means that the `certbot-apache` plugin is missing. Since Bitnami uses a custom Apache setup, Certbot's Apache plugin does not work properly with it. You need to use the **standalone method** instead.

Solution: Use Certbot Standalone Mode

Since Bitnami doesn't use the standard Apache stru e, you should use **Certbot with the standalone mode** instead of the Apache plugin.

Step 1: Stop Apache

First, you need to stop Apache temporarily to allow Certbot to use port 80 for verification.

```bash
sudo /opt/bitnami/ctlscript.sh stop apache
```

Step 2: Run Certbot in Standalone Mode

Run this command to generate an SSL certificate:

```bash
sudo certbot certonly --standalone -d nafiulakbar.xyz -d backend.nafiulakbar.xyz
```

- This will prompt you for your email and agree to the terms.
- Certbot will verify the domain ownership and generate certificates.

Step 3: Restart Apache

Once the SSL certificate is generated, restart Apache:

```bash
sudo /opt/bitnami/ctlscript.sh start apache
```

Step 4: Configure Apache for SSL

Now, update your virtual hosts with the correct SSL paths.

1 Edit `nafiulakbar-vhost-ssl.conf`

```bash
sudo nano /opt/bitnami/apache2/conf/vhosts/nafiulakbar-vhost-ssl.conf
```

Update it with:

```apache
<VirtualHost *:443>
    ServerName nafiulakbar.xyz
    DocumentRoot "/opt/bitnami/htdocs/nafiulakbar"

    SSLEngine on
    SSLCertificateFile "/etc/letsencrypt/live/nafiulakbar.xyz/fullchain.pem"
    SSLCertificateKeyFile "/etc/letsencrypt/live/nafiulakbar.xyz/privkey.pem"

    <Directory "/opt/bitnami/htdocs/nafiulakbar">
        AllowOverride All
        Require all granted
    </Directory>

    ErrorLog "logs/nafiulakbar-ssl-error_log"
    CustomLog "logs/nafiulakbar-ssl-access_log" common
</VirtualHost>
```

2 Edit `backend-nafiulakbar-vhost-ssl.conf`

```bash
sudo nano /opt/bitnami/apache2/conf/vhosts/backend-nafiulakbar-vhost-ssl.conf
```

Update it with:

```apache
```

```
<VirtualHost *:443>
    ServerName backend.nafiulakbar.xyz
    DocumentRoot "/opt/bitnami/htdocs/backend"

    SSLEngine on
    SSLCertificateFile "/etc/letsencrypt/live/nafiulakbar.xyz/fullchain.pem"
    SSLCertificateKeyFile "/etc/letsencrypt/live/nafiulakbar.xyz/privkey.pem"

    <Directory "/opt/bitnami/htdocs/backend">
        AllowOverride All
        Require all granted
    </Directory>

    ErrorLog "logs/backend-ssl-error_log"
    CustomLog "logs/backend-ssl-access_log" common
</VirtualHost>
```

Step 5: Restart Apache

bash Copy Edit

```
sudo /opt/bitnami/ctlscript.sh restart apache
```

Step 6: Set Up SSL Auto-Renewal

Let's Encrypt certificates expire every 90 days, so yo ↓ ed to set up auto-renewal.

Edit the crontab:

bash Copy Edit

```
sudo crontab -e
```

Add this line at the bottom:

bash Copy Edit

```
0 3 * * * /usr/bin/certbot renew --quiet --standalone --pre-hook "sudo /opt/bitnami/ctlscript
```

This will:

- Stop Apache
- Renew the certificate
- Restart Apache
  It runs **every day at 3 AM**.

Step 7: Verify SSL

1. Run:

   bash Copy Edit

   ```
   sudo certbot renew --dry-run
   ```

   This simulates the renewal process to ensure everything is working.
2. Visit your sites:
   - ✅ `https://nafiulakbar.xyz`
   - ✅ `https://backend.nafiulakbar.xyz`

Now, both your main domain and subdomain should be secured with Let's Encrypt SSL. 🚀 Let me know if you face any issues!

After runnig sudo certbot renew --dry-run

It says

Processing /etc/letsencrypt/renewal/nafiulakbar.xyz.conf
- -
Account registered.
Simulating renewal of an existing certificate for nafiulakbar.xyz and backend.nafiulakbar.xyz
Failed to renew certificate nafiulakbar.xyz with error: Could not bind TCP port 80 because it is already in use by another process on this systeserver). Please stop the program in question and then try again.

- -
All simulated renewals failed. The following certificates could not be renewed:
/etc/letsencrypt/live/nafiulakbar.xyz/fullchain.pem (failure)
- -
1 renew failure(s), 0 parse failure(s)

The error message indicates that **port 80 is already in use**, which prevents Certbot from binding to it for the renewal process. This happens because Apache (Bitnami) is running and using port 80.

Solution: Stop Apache Before Renewal

Since Let's Encrypt needs port 80 to verify the certificate, you must stop Apache before running the renewal command.

Step 1: Stop Apache

Run:

```bash
sudo /opt/bitnami/ctlscript.sh stop apache
```

Step 2: Run Certificate Renewal

Manually renew the certificates:

```bash
sudo certbot renew --standalone
```

Step 3: Restart Apache

Once the renewal is complete, restart Apache:

```bash
sudo /opt/bitnami/ctlscript.sh start apache
```

Step 4: Automate Renewal with Cron Job

To avoid manually stopping and starting Apache ev time, set up a cron job that does it automatically.

Edit the crontab:

```bash
sudo crontab -e
```

Add this line at the bottom:

```bash
0 3 * * * /usr/bin/certbot renew --quiet --standalone --pre-hook "sudo /opt/bitnami/ctlscript
```

This will:

1. Stop Apache
2. Renew the certificate
3. Restart Apache
   ⏳ It runs **every day at 3 AM** automatically.

Step 5: Test Auto-Renewal

Run:

```bash
sudo certbot renew --dry-run
```

This will simulate the renewal process to ensure it works without errors.

Now, your SSL certificate should renew automatically without manual intervention. Let me know if you face any issues! 🚀

Now I have another problem, when I visit https://nafiulakbar.xyz/ or https://backend.nafiulakbar.xyz/. the page loads but it shows,

Forbidden
You don't have permission to access this resource.

The **"Forbidden – You don't have permission to access this resource"** error usually happens due to **Apache permission issues** or **misconfigured Virtual Hosts** in Bitnami. Let's fix it step by step.

Step 1: Check Apache Virtual Host Configuration

Run the following command to open your virtual host configuration:

```bash
sudo nano /opt/bitnami/apache2/conf/vhosts/nafiulakbar-vhost-ssl.conf
```

Make sure it contains something like this:

```apache
<VirtualHost *:443>
    ServerName nafiulakbar.xyz
    DocumentRoot "/opt/bitnami/htdocs/nafiulakbar"

    SSLEngine on
    SSLCertificateFile "/etc/letsencrypt/live/nafiulakbar.xyz/fullchain.pem"
```

```
    SSLCertificateKeyFile "/etc/letsencrypt/live/nafiulakbar.xyz/privkey.pem"

    <Directory "/opt/bitnami/htdocs/nafiulakbar">
        AllowOverride All
        Require all granted
    </Directory>

    ErrorLog "logs/nafiulakbar-ssl-error_log"
    CustomLog "logs/nafiulakbar-ssl-access_log" common
</VirtualHost>
```

💡 **Make sure:**

- The **DocumentRoot** is correct (where your Laravel project is located).
- The `<Directory>` block includes:

```apache
AllowOverride All
Require all granted
```

Now do the same for the backend:

```bash
sudo nano /opt/bitnami/apache2/conf/vhosts/backend-nafiulakbar-vhost-ssl.conf
```

Make sure it looks like this:

```apache
<VirtualHost *:443>
    ServerName backend.nafiulakbar.xyz
    DocumentRoot "/opt/bitnami/htdocs/backend"

    SSLEngine on
    SSLCertificateFile "/etc/letsencrypt/live/nafiulakbar.xyz/fullchain.pem"
    SSLCertificateKeyFile "/etc/letsencrypt/live/nafiulakbar.xyz/privkey.pem"

    <Directory "/opt/bitnami/htdocs/backend">
        AllowOverride All
        Require all granted
    </Directory>

    ErrorLog "logs/backend-ssl-error_log"
    CustomLog "logs/backend-ssl-access_log" common
</VirtualHost>
```

Step 2: Check Folder Permissions

Run the following commands to make sure the Laravel project has correct permissions:

```bash
sudo chown -R bitnami:daemon /opt/bitnami/htdocs/nafiulakbar
sudo chown -R bitnami:daemon /opt/bitnami/htdocs/backend

sudo find /opt/bitnami/htdocs/nafiulakbar -type d -exec chmod 755 {} \;
sudo find /opt/bitnami/htdocs/nafiulakbar -type f -exec chmod 644 {} \;

sudo find /opt/bitnami/htdocs/backend -type d -exec chmod 755 {} \;
sudo find /opt/bitnami/htdocs/backend -type f -exec chmod 644 {} \;

sudo chmod -R 775 /opt/bitnami/htdocs/nafiulakbar/storage
sudo chmod -R 775 /opt/bitnami/htdocs/backend/storage
```

This ensures: ✅ Laravel folders are owned by **bitnami**
✅ Directories have **755** permissions
✅ Files have **644** permissions
✅ Storage folders have **775** permissions

Step 3: Restart Apache

After making these changes, restart Apache:

```bash
sudo /opt/bitnami/ctlscript.sh restart apache
```

Step 4: Check Laravel `.htaccess` File

Bitnami disables `.htaccess` by default, so create a custom one.

Edit:

```bash
sudo nano /opt/bitnami/apache2/conf/vhosts/htaccess/nafiulakbar-htaccess.conf
```

Add:

```apache
<Directory "/opt/bitnami/htdocs/nafiulakbar/public">
    AllowOverride All
</Directory>
```

Now, create a similar file for the backend:

```bash
```

```
sudo nano /opt/bitnami/apache2/conf/vhosts/htaccess/backend-nafiulakbar-htaccess.conf
```

Add:

```apache
<Directory "/opt/bitnami/htdocs/backend/public">
    AllowOverride All
</Directory>
```

Now restart Apache again:

```bash
sudo /opt/bitnami/ctlscript.sh restart apache
```

Step 5: Set the Correct Laravel Public Directory

Make sure Apache is pointing to the **public** directory of Laravel.

Edit your virtual hosts again:

```bash
sudo nano /opt/bitnami/apache2/conf/vhosts/nafiulakbar-vhost-ssl.conf
```

Change:

```apache
DocumentRoot "/opt/bitnami/htdocs/nafiulakbar/public"
<Directory "/opt/bitnami/htdocs/nafiulakbar/public">
```

Do the same for the backend:

```bash
sudo nano /opt/bitnami/apache2/conf/vhosts/backend-nafiulakbar-vhost-ssl.conf
```

Change:

```apache
DocumentRoot "/opt/bitnami/htdocs/backend/public"
<Directory "/opt/bitnami/htdocs/backend/public">
```

Restart Apache again:

```bash
sudo /opt/bitnami/ctlscript.sh restart apache
```

Now, visit:

- `https://nafiulakbar.xyz`
- `https://backend.nafiulakbar.xyz`

Everything should work! 🚀 Let me know if you still face issues.

Ask anything
Search
Reason
ChatGPT can make mistakes. Check important info.